

Jia Yi Ooi

Senior Full-Stack Engineer

Penang, Malaysia (UTC+8) · Remote worldwide or relocation · me@jyooi.com

jyooi.com · [LinkedIn](#) · [GitHub](#)

SUMMARY

Full-stack engineer with 9 years shipping products end to end, the last 5+ fully remote for teams in Singapore, the US, and Australia. Most recently built and operated the messaging platform behind 500+ companies and 100k+ messages a day. Owns both the user-facing product and the real-time services behind it, from empty repo to launch. Works daily with coding agents such as Claude Code and pi, and builds the agentic workflows and tooling around them. Available worldwide as a contractor through my own registered entity, or as a direct hire with relocation.

LANGUAGES & TECHNOLOGIES

Languages: TypeScript, Python, Elixir, SQL
Frontend: React, Next.js, Vite, TanStack Router, React Query, Socket.IO, Electron, Playwright, Vitest
Backend: Node.js, FastAPI, PostgreSQL, REST API design, load testing
Infrastructure: Kubernetes, Docker, CI/CD (GitHub Actions), Linux, Git
Other: Design systems, real-time systems, Web3 wallets (MetaMask, Tezos)

EXPERIENCE

Career development sabbatical

Apr 2026 – Present

- Built [elixir-opentui](#), [agent-simple-english](#), and [pi-ask-user-question](#) - see Projects.

Uchat

Nov 2024 – Apr 2026

Senior Software Engineer, Business unit

Remote (Singapore)

Business platform of Uchat, a consumer messaging app in Thailand. 500+ companies run their B2C conversations on it.

- Shipped the real-time platform processing 100k+ messages a day for 500+ business accounts - React/TypeScript (Vite, TanStack Router, Socket.IO) over FastAPI on Kubernetes.
- Built the strongly-typed UI component system the platform is assembled from, giving the team a shared vocabulary for new screens.
- Load-tested the chat service to 100k concurrent connections ahead of traffic growth.
- Reviewed the team's frontend work and shaped feature scope directly with product and design.

Accomy

Feb 2024 – Oct 2024

Senior Software Engineer

Remote (Singapore)

- Led 2 frontend engineers to ship the MVP of a corporate hotel booking platform in 4 months, from empty repo to launch.
- Established the Next.js + TypeScript codebase from scratch, mentored 2 engineers onto it, and drove architecture decisions with product and design.

Coinalpha (Hummingbot)

Jun 2021 – Nov 2023

Software Engineer

Remote (US)

Hummingbot is an open-source market-making framework for crypto trading bots.

- Built the React + FastAPI liquidity mining platform used by ~1,000 bot operators facilitating ~\$100k in daily trading volume.
- Built and published [@hummingbot/hbui](#), the Hummingbot design system - design tokens synced from Figma into a Tailwind + React package consumed across the organisation's frontends.
- Fixed memory leaks that exhausted 8 GB machines within hours, cutting steady-state memory ~90%.

Areca System

Jul 2020 – May 2021

Contract Software Engineer

Penang, Malaysia

- Built an Electron point-of-sale system from scratch for a 30-store supermarket chain.

Aglive

Nov 2017 – Jun 2020

Software Engineer

Remote (Australia)

- Built and operated the tracking dashboard covering 500+ trucks and buses on the Austracker platform.
- Led deduplication of multi-million-row GPS data, removing ~25% duplicate records that fed customer reports.

PROJECTS

- [elixir-opentui](#) - terminal UI framework for Elixir with Zig NIF backend: Elm-inspired architecture, declarative View DSL, 15+ widgets, double-buffered rendering.
- [agent-simple-english](#) - deterministic ASD-STE100 linter: Engine, CLI, and pi/Claude Code adapters that gate writes, edits, commits, and replies.
- [pi-ask-user-question](#) - pi TUI extension for structured questions: multi-select, free-text Other, TUI-only with transcript rendering.

EDUCATION

B.IT Communications & Networking (Hons), Universiti Tunku Abdul Rahman

May 2017